1.INTRODUCTION TO PYTHON PACKAGES

Python packages are structured collections of modules that allow developers to organize code efficiently and reuse functionalities across multiple projects. A package is essentially a directory containing multiple Python files along with a special **__init__.py** file.

Packages play a crucial role in:

- **Code reusability**
- **Modular programming**
- **Improved readability**
- **Scalability of applications**

In real-world software development, packages help break large programs into smaller, manageable components.

OBJECTIVE OF THIS ASSIGNMENT

This assignment aims to explore and implement three fundamental Python standard library packages:

- **random** → Used for generating random values
- **math** → Provides mathematical functions and constants
- **datetime** → Handles date and time operations

Each package is demonstrated using multiple programs ranging from basic usage to real-world applications.

WHY THESE PACKAGES

random

The **random** module is widely used in:

- Game development (dice, cards)
- Simulations (Monte Carlo methods)
- Data sampling
- AI randomness & testing

It uses a **pseudo-random number generator (Mersenne Twister)** which is deterministic but appears random.

math

The **math** module is essential for:

- Scientific calculations
- Engineering problems
- Geometry
- Trigonometry
- Logarithmic computations

It provides high-performance functions implemented in C.

datetime

The **datetime** module is used in:

- Scheduling systems
- Age calculation
- Logging systems
- Event tracking
- Time-based automation

2.PACKAGE 1-random

The **random** module generates pseudo-random numbers. These numbers are not truly random but are generated using algorithms.

Important Features:

- Reproducibility using **seed()**
- Sampling with/without replacement
- Uniform and weighted distributions

PROGRAM 1: Advanced Random Generator Toolkit

SOURCE CODE:

main.py

```python
import random
print("RANDOM TOOLKIT\n")
print("Random Float:", random.random())
print("Random Integer (1-100):", random.randint(1, 100))
items = ["Pen", "Book", "Laptop", "Phone"]
print("Random Item:", random.choice(items))
random.shuffle(items)
print("Shuffled Items:", items)
print("Sample:", random.sample(range(1, 50), 5))
print("Weighted Choice:", random.choices(["A", "B", "C"], weights=[1,
    3, 6], k=5))
```

OUTPUT:

```
Output                                                    Clear
RANDOM TOOLKIT

Random Float: 0.738202402855079
Random Integer (1-100): 10
Random Item: Pen
Shuffled Items: ['Laptop', 'Pen', 'Book', 'Phone']
Sample: [27, 8, 6, 41, 46]
Weighted Choice: ['C', 'C', 'B', 'C', 'C']

=== Code Execution Successful ===
```

PROGRAM 2: Password Generator

SOURCE CODE:

```python
main.py                                    Share  Run
import random
import string

length = 10

characters = string.ascii_letters + string.digits + "!@#$%"

password = ''.join(random.choice(characters) for _ in range(length))

print("Generated Password:", password)
```

OUTPUT:

Output

```
Generated Password: %Upq2wgYUR

=== Code Execution Successful ===
```

PROGRAM 3: Dice Simulation (1000 Rolls)

SOURCE CODE:

main.py

```python
import random

results = {i:0 for i in range(1,7)}

for _ in range(1000):
    roll = random.randint(1,6)
    results[roll] += 1

print("Dice Roll Frequency:")
for k,v in results.items():
    print(k, ":", v)
```

OUTPUT:

```
Output                                                    Clear
Dice Roll Frequency:
1 : 162
2 : 172
3 : 166
4 : 160
5 : 159
6 : 181

=== Code Execution Successful ===
```

3. PACKAGE 2 — math

The **math** module provides access to advanced mathematical operations.

Key Categories:

- Constants → π, e
- Algebra → sqrt, power
- Trigonometry → sin, cos, tan
- Logarithms → log, exp
- Number theory → gcd, factorial

PROGRAM 1: Scientific Calculator

SOURCE CODE:

main.py

```python
import math

num = 25

print("Square Root:", math.sqrt(num))
print("Power (2^5):", math.pow(2,5))
print("Factorial:", math.factorial(5))
print("Log (base e):", math.log(10))
print("Log base 10:", math.log10(1000))
```

OUTPUT:

Output Clear

```
Square Root: 5.0
Power (2^5): 32.0
Factorial: 120
Log (base e): 2.302585092994046
Log base 10: 3.0

=== Code Execution Successful ===
```

PROGRAM 2: Trigonometry Table

SOURCE CODE:

main.py Share Run

```python
import math

for angle in [0, 30, 45, 60, 90]:
    rad = math.radians(angle)
    print(f"Angle: {angle}")
    print("sin:", math.sin(rad))
    print("cos:", math.cos(rad))
    print("tan:", math.tan(rad))
    print()
```

OUTPUT:

Output Clear

```
Angle: 0
sin: 0.0
cos: 1.0
tan: 0.0

Angle: 30
sin: 0.49999999999999994
cos: 0.8660254037844387
tan: 0.5773502691896257

Angle: 45
sin: 0.7071067811865475
cos: 0.7071067811865476
tan: 0.9999999999999999

Angle: 60
sin: 0.8660254037844386
cos: 0.5000000000000001
tan: 1.7320508075688767

Angle: 90
sin: 1.0
cos: 6.123233995736766e-17
tan: 1.633123935319537e+16
```

PROGRAM 3: Geometry Solver

SOURCE CODE:

main.py

```python
import math

r = 5

area = math.pi * r**2
circumference = 2 * math.pi * r

print("Circle Area:", area)
print("Circumference:", circumference)
```

OUTPUT:

Output

```
Circle Area: 78.53981633974483
Circumference: 31.41592653589793

=== Code Execution Successful ===
```

4. PACKAGE 3 — datetime

The **datetime** module provides classes for working with dates and time.

Core Classes:

- **date**
- **time**
- **datetime**
- **Timedelta**

PROGRAM 1: Current Date & Time Info

SOURCE CODE:

main.py

```python
from datetime import datetime

now = datetime.now()

print("Current Date & Time:", now)
print("Year:", now.year)
print("Month:", now.month)
print("Day:", now.day)
```

OUTPUT:

Output Clear

```
Current Date & Time: 2026-05-06 22:34:44.633049
Year: 2026
Month: 5
Day: 6

=== Code Execution Successful ===
```

PROGRAM 2: Age Calculator

SOURCE CODE:

main.py Share Run

```python
from datetime import date

birth_year = 2005
birth_month = 5
birth_day = 10

today = date.today()

age = today.year - birth_year

print("Your Age:", age)
```

OUTPUT:

```
Output

Your Age: 21

=== Code Execution Successful ===
```

PROGRAM 3: Countdown Timer

SOURCE CODE:

main.py

```python
from datetime import datetime

event_date = datetime(2026, 1, 1)
today = datetime.now()

remaining = event_date - today

print("Days Remaining:", remaining.days)
```

OUTPUT:

```
Output                                                    Clear
Days Remaining: -126

=== Code Execution Successful ===
```

5. CONCLUSION & LEARNINGS

This assignment provided a deep understanding of Python’s standard library packages.

Key Takeaways:

- **random** helps simulate real-world uncertainty
- **math** enables complex calculations efficiently
- **datetime** simplifies time-based operations